

What you'll *actually* do

The question that comes up in every information session: not what LENS believes, but what a student does. This is the work — one capability problem carried end to end, and the artifact you produce at each step.

LENS is not another way to build a course. The unit of work is the *system*, and the deliverable is *evidence*.

You still design. But you design against a capability requirement decomposed from what the system has to deliver, you instrument what you build, and you report what the evidence forced you to change. The first move of the work is deciding whether a learning intervention is the right answer at all.

ONE PROBLEM, END TO END

THE STEP	WHAT YOU DO	WHAT YOU PRODUCE
1 Frame it	Sit with the people doing the work. Separate the complaint from the failure, and restate the request as a system performance requirement rather than a training order.	Problem framing with stakeholder and context analysis LEN 6
2 Model the system	Diagram the learning environment and the operational system that hosts it — components, interfaces, feedback loops — and place the capability decisions on the system's lifecycle.	Capability analysis and system model LEN 1 · ASSIGNMENT 1
3 Attribute the gap	Decide whether the shortfall sits in what people can do, in how the system is designed, or in how the organization runs — and name the evidence that would change your answer.	Gap attribution with its evidence basis LEN 4
4 Design it	Build the smallest thing that could work — including the human-machine boundary: who holds which role, what the system decides, and what evidence takes a delegation back.	Role architecture and delegation-with-revocation spec LEN 2
5 Instrument it	Measure ethically, at the speed the operation actually moves, and measure the team as a unit rather than only the individual operator. State what you cannot see.	Instrumentation plan and evidence dashboard LEN 4 · LEN 10
6 Run the cycle	Design, instrument, evaluate, refine — then report, in first person, the framing the evidence forced you to revise. An iteration that revised nothing has to argue why.	Iteration cycle report LEN 1 · ASSIGNMENT 2
7 Make it survive	Plan adoption and sustainment against the cost of owning the thing, not the cost of building it, and design the support layer so it runs when you are not in the room.	Implementation and sustainment plan, job aids LEN 6 · LEN 10
8 Defend it	Brief the people who will act on it — technical and non-technical in the same room — stating the uncertainty and the limits of what you can infer.	Reproducible report, governance plan, 12-minute briefing LEN 1 · LEN 10

THE SIX LENS COURSES, AND WHERE THE ELECTIVES GO

LEN 1 · Principles of Learning Engineering for Complex Systems

LEN 2 · Human-Machine Teaming and Adaptive Learning Systems

LEN 3 · Learning Engineering Systems

LEN 6 · Applied Problem Solving in Learning Engineering

LEN 4 · Evidence, Analytics, and Measurement

LEN 10 · Learning Engineering Project (capstone)

Electives · Human Capability Analysis · Computational and AI Methods
· Bias, Risk & Governance · Knowledge Transfer

EVERY STEP IS GRADED WORK

These are the actual assignments, not a metaphor for them. LEN 1 alone asks for a capability analysis, a documented iteration cycle, a coupling-failure analysis, and a recorded briefing.

YOUR DOMAIN IS THE MATERIAL

Assignments run on your own professional setting wherever they can. Prior expertise is treated as an asset to work from, not a gap to remediate.

THE CAPSTONE IS THE WHOLE ARC

LEN 10 runs steps 1 through 8 at a partner organization on a real capability problem, with at least one complete iteration cycle documented.

Coming across from *LXD or AILE*

If you come from learning experience design

CARRIES OVER The learning sciences, needs analysis, and design craft. LENS assumes them — the Learning Sciences Studio is a shared foundation course, and production of content and media stays with LXD.

WHAT'S NEW The unit of analysis moves outward: from the experience to the operational system that hosts it. You write capability requirements decomposed from a system requirement, and you measure against operational outcomes rather than completion and satisfaction.

THE MOVE YOU'LL LEARN Telling a learning problem from a system problem — and being willing to report that the intervention you were asked for is not the fix.

If you come from AI leadership in education

CARRIES OVER Fluency with the tools, and the governance, adoption, and organizational questions that come with them. LENS treats AI as one case of human-system collaboration, not a separate subject.

WHAT'S NEW You specify the partnership as an engineered object: role architecture, authority gradients, mode and state transparency, recoverability — and delegation with revocation, where you name in advance the evidence that would take the delegation back.

THE MOVE YOU'LL LEARN Measuring the human-machine team as a unit of analysis, and specifying capability requirements for systems that have not been fielded yet, from design artifacts alone.

WHAT CHANGES BETWEEN ARRIVING AND LEAVING

YOU ARRIVE ABLE TO

Attribute a performance problem to training by default.

Interpret simple descriptive data about learners — completion, scores — as someone else presents it.

Recognize that learning solutions sit inside larger organizational and technical systems.

Identify stakeholders in a familiar setting and describe their needs informally.

Use AI tools as a consumer.

Work inside your own professional silo, handing off at its boundary.

YOU LEAVE ABLE TO

Diagnose whether the gap is rooted in learning, in system design, or in organizational conditions — before proposing anything.

Instrument a learning environment ethically and produce decision-grade evidence: dashboards and reproducible reports linking learning measures to operational performance.

Model the learning environment and its host system, translating capability requirements across the human-machine interface into design requirements.

Run a stakeholder and context analysis in a high-consequence environment using a human systems integration framework, producing a documented context model.

Design capability development for human-AI teams and specify learning requirements for systems not yet fielded.

Lead and integrate a learning engineering team across disciplines, with the shared vocabulary the work requires.

WHERE THE VALUE LANDS

Money not spent on the wrong fix

Training is the default answer to a performance problem, and often the wrong one. A graduate who can attribute the gap — with evidence — stops the spend that was never going to work.

Evidence a decision-maker can act on

Not completion rates. Dashboards, reproducible reports, and governance documentation that state what is known, what is assumed, and what would change the decision.

Capability specified before it is late

Requirements written from design artifacts, before a system is fielded — so capability is designed into the lifecycle rather than bolted on when training arrives last.

ASKED IN EVERY SESSION

Do I need to code?

No. Computational and AI methods are a recommended elective, not a gate. The required methods courses teach you to produce and appraise evidence, and to brief the specialist where depth is needed.

Do I have to work in healthcare or defense?

No. Those are two of seven representative domains, and the list is not exhaustive. Education is on it as a domain and is also the discipline LENS works from.

Is this a research degree?

No. Study design and human-subjects depth are treated at awareness and routed to an elective. The capstone is applied work in a real organization.